egg's constituents become more and more disordered—that the egg splatters rather than unsplatters, thereby also achieving higher entropy.

As you can see, the concept of entropy provides a precise version of the "easy versus difficult" conclusion we found earlier. It's easy for the pages of *War and Peace* to fall out of order because there are *so many* out-of-order arrangements. It's difficult for the pages to fall in perfect order because hundreds of pages would need to move in just the right way to land in the unique sequence Tolstoy intended. It's easy for an egg to splatter because there are *so* many ways to splatter. It's difficult for an egg to unsplatter, because an enormous number of splattered constituents must move in perfect coordination to produce the single, unique result of a pristine egg resting on the counter. For things with many constituents, going from lower to higher entropy—from order to disorder—is easy, so it happens all the time. Going from higher to lower entropy—from disorder to order—is harder, so it happens rarely, at best.

Notice, too, that this entropic arrow is not completely rigid; there is no claim that this definition of time's direction is 100 percent foolproof. Instead, the approach has enough flexibility to allow these and other processes to happen in reverse as well. Since the second law proclaims that entropy increase is only a statistical likelihood, not an inviolable fact of nature, it allows for the rare possibility that pages can fall into perfect numerical order, that gas molecules can coalesce and reenter a bottle, and that eggs can unsplatter. By using the mathematics of entropy, the second law expresses precisely how statistically unlikely these events are (remember, the huge number on this page–this page reflects how much more likely it is that pages will land out of order), but it recognizes that they can happen.

This seems like a convincing story. Statistical and probabilistic reasoning has given us the second law of thermodynamics. In turn, the second law has provided us with an intuitive distinction between what we call past and what we call future. It has given us a practical explanation for why things in daily life, things that are typically composed of huge numbers of constituents, start like *this* and end like *that*, while we never see them start like *that* and end like *this*. But over the course of many years— and thanks to important contributions by physicists like Lord Kelvin, Josef Loschmidt, Henri Poincaré, S. H.